PHAM MINH NHAT

SOFTWARE ENGINEER

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SUMMARY

Software Engineer with production experience across backend systems, cloud infrastructure, and full-stack delivery. Built and shipped real systems at Grevo Solutions and contributed to AWS infrastructure design during an internship at Amazon. Comfortable owning a feature end-to-end from system design to deployment.

TECHNICAL SKILLS

- **Programming Languages:**
  - JavaScript, Java, Python
- **Framework/ Platforms:**
  - ReactJS, NodeJS, ThreeJS
  - Spring Boot
- **Database:**
  - MySQL, PostgreSQL
  - Redis
- **DevOps & Infra:**
  - Git, Docker, Jenkins
  - Google Cloud Run
  - Render, Vercel, Railway
  - AWS (EC2, S3, Lambda)
- **Foreign Languages:**
  - English
  - Japanese (basic)
- **Others**
  - Strong understanding of OOP, SOLID principles, and RESTful API design.
  - Hands-on experience with CI/CD pipelines (Jenkins) and containerized deployments (Docker, Google Cloud Run).
  - Familiar with AWS Well-Architected Framework across security, reliability, and performance pillars.

WORK EXPERIENCE

Amazon Web Services (AWS)
Cloud Infrastructure Intern

Jan 2026 - Aug 2026
Ho Chi Minh City · On-site

- **Description:**
  - Contributed to designing and optimizing cloud infrastructure for a subscription-based web application on AWS, from architecture evaluation through hands-on implementation and cost optimization.
- **Technologies:**
  - EC2, S3, CloudFront, RDS, Lambda, API Gateway, ElastiCache (Redis), ALB, CloudWatch
- **Responsibilities:**
  - Researched and recommended AWS architecture patterns tailored to workload profiles of thousands to tens of thousands of daily requests.
  - Participated in architectural decisions and hands-on implementation of a subscription-based web application.
  - Implemented monitoring and observability using CloudWatch for faster detection of performance bottlenecks.
  - Optimized EC2 sizing, caching strategy, and storage configuration, contributing to ~17% cloud cost reduction.

Grevo Solutions
Founding Engineer & Project Lead

Mar 2025 - Jan 2026
Ho Chi Minh City · On-site

- **Description:**
  - Led end-to-end technical ownership of a production-grade urban waste management platform, driving architecture decisions and team delivery from system design through cloud deployment.
- **Technologies:**
  - Java, Spring Boot, PostgreSQL, Redis, WebSocket, Docker, Jenkins, Google Cloud Run, AWS S3, VietMap API
- **Responsibilities:**
  - Led a team of 4 engineers across backend, frontend, and cloud infrastructure, owning architecture, code quality, and end-to-end delivery.
  - Designed and implemented a Spring Boot backend with 91 REST APIs supporting citizen, collector, enterprise, and admin workflows.
  - Built a Grab-inspired intelligent dispatch engine with GPS-based collector matching, coordinate smoothing, and dynamic order batching/splitting, reducing manual coordination by ~40%.
  - Built a WebSocket-based real-time communication layer for task assignment, status updates, and live tracking.
  - Implemented production-grade security and quality standards with JWT, OAuth2, RBAC, and over 81% unit/integration test coverage.
  - Managed CI/CD via Jenkins, containerized deployments on Google Cloud Run, and migrated media storage to Amazon S3 for improved reliability and maintainability.

FPT Education
Technical Staff

Jan 2023 – Aug 2024
Ho Chi Minh City · On-site

- **Description:**
  - Supported technical operations and internal software development at FPT Education, while contributing to organizational events and student-facing initiatives.
- **Responsibilities:**
  - Built and maintained an internal student management web application serving the FPT Education division, handling user account management and reporting workflows.
  - Managed and maintained computer equipment and technical facilities to ensure system availability.
  - Coordinated major events including F-Camp 2023; recognized as one of the top 20 outstanding contributors in the organization.

EDUCATION

FPT University
Bachelor of Engineering · Computer Science

Sep 2022 – Present
Ho Chi Minh City

Multimedia University
Student Exchange Program · Interpersonal Communication

Jan 2023 – Apr 2023
Malaysia

RELEVANT PROJECTS

LordNine Bot · MMORPG Automation Framework Mar 2026 – Apr 2026
Personal Project

- **Description:**
  - Built and shipped a production-grade MMORPG automation framework as a SaaS product, handling 100+ concurrent instances with a full distribution pipeline iterated across 34 releases in 2 months.
- **Technologies:**
  - Python 3.13, OpenCV, Chrome DevTools Protocol, CustomTkinter, SQLite, Vercel, PyInstaller, Inno Setup
- **Responsibilities:**
  - Built a production-grade NPC automation engine controlling 100+ concurrent instances with sessions up to 36 hours.
  - Designed a no-code visual flowchart engine achieving ~85% task completion rate and 99.99% visual state detection accuracy via OpenCV template matching.
  - Replaced Playwright with a custom CDP client, reducing per-instance memory from ~100MB to ~1KB – enabling 100+ concurrency on consumer hardware.
  - Implemented AIMD adaptive screenshot scheduling (150ms–800ms) and GUI queue coalescing, achieving ~90% reduction in UI update overhead.
  - Shipped a complete SaaS pipeline: HWID-bound license system, SHA256-verified auto-update, and Windows installer – iterated to v3.24+ across 34 releases in 2 months.

Attenda · Smart Attendance System Mar 2025 – Aug 2025
Team Lead

- **Description:**
  - Led a 4-person team to build a cost-optimized smart attendance system for university use, combining ESP32-S3 edge hardware with a cloud-connected web dashboard.
- **Technologies:**
  - ESP32-S3, TensorFlow, React TypeScript, Python, Gunicorn, PostgreSQL
- **Responsibilities:**
  - Engineered edge-based attendance hardware using ESP32-S3 with fingerprint sensor, minimizing cloud overhead through local processing; implemented TensorFlow face recognition achieving ~95% accuracy with <1.5s latency.
  - Built a React + TypeScript dashboard managing 1,000+ student records with real-time monitoring; designed Python/PostgreSQL backend handling 500+ daily attendance events.